

Log-normal Distribution:

The positive random variable X is said to have a log-normal distribution; if $\log_e X$ is normally distributed.

$$\text{Let } Y = \log_e X \sim N(\mu, \sigma^2), \text{ for } x > 0$$

$$F_X(x) = P(X \leq x) = P(\log_e X \leq \log_e x) = P(Y \leq \log_e x)$$

$$= \int_{-\infty}^{\log_e x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}} dy \quad (\text{Since } \log_e x \text{ is monotonic increasing function})$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_0^x \exp\left[-\frac{1}{2}\frac{(\log u - \mu)^2}{\sigma^2}\right] \frac{du}{u}, \quad \begin{matrix} y = \log u \\ \frac{dy}{du} = \frac{1}{u} \end{matrix}$$

For $x \leq 0$, $F_X(x) = P(X \leq x) = 0$ because X is a positive r.v.

$$\text{Let us define } f_X(u) = \frac{1}{u\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\log u - \mu)^2}, \quad u > 0 \quad \text{--- (1)}$$

$$= 0, \quad \text{if } u \leq 0$$

Then $F_X(x) = \int_{-\infty}^x f_X(u) du$ for $x > 0$ and hence $f(x)$ defined in (1) is a pdf of X .

Remark: If $\log X \sim N(\mu, \sigma^2)$, then $X = e^Y$, is called a log-normal random variable, since its logarithm $\log X = Y$ is a normal r.v.

Moments: The r th moment about origin is given by

$$\begin{aligned} \mu_r' &= E(X^r) = E(e^{rY}) \quad [\because Y = \log X \Rightarrow X = e^Y] \\ &= M_Y(r) \quad (\text{M.g.f of } Y, r \text{ being the real const (parameter)}) \\ &= \exp\left[\mu r + \frac{1}{2}r^2\sigma^2\right] \quad (\because Y \sim N(\mu, \sigma^2)) \end{aligned}$$

Remarks: 1. In particular if we take $\mu = \log d$, $d > 0$. i.e. $\log X \sim N(\log d, \sigma^2)$, then

$$\mu_r' = E(X^r) = \exp\left(r \log d + \frac{1}{2}r^2\sigma^2\right) = d^r \exp\left(\frac{r^2\sigma^2}{2}\right)$$

$$\text{Mean } \mu_1' = d e^{\frac{\sigma^2}{2}} \quad \text{and } \mu_2' - \mu_1'^2 = d^2 e^{\sigma^2} (e^{\sigma^2} - 1)$$

2. Log normal distⁿ arises in problems of economics, biology, geology and reliability theory. In particular, it arises in the study of dimensions of particles under pulverisation.

Exponential Prob Distribution (Cont Prob distⁿ)

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Defⁿ: A random variable X is said to have an exponential distⁿ with parameter $\theta > 0$, if its pdf is given by:

$$f(x, \theta) = \theta e^{-\theta x} ; x \geq 0$$

Distⁿ Function $F_X(x) = P(X \leq x) = \int_0^x \theta e^{-\theta x} dx = \theta \left[\frac{-e^{-\theta x}}{\theta} \right]_0^x$

$$\boxed{F_X(x) = 1 - e^{-\theta x} ; x \geq 0}$$

MGF of Exp. prob distⁿ

$$\begin{aligned} M_X(t) &= E e^{tx} = \int_0^\infty e^{tx} f(x) dx = \int_0^\infty e^{tx} \theta e^{-\theta x} dx \\ &= \theta \int_0^\infty e^{-(\theta-t)x} dx = \frac{\theta}{(\theta-t)} , \theta > t \\ &= \frac{\theta}{\theta(1-\frac{t}{\theta})} = (1-\frac{t}{\theta})^{-1} = \sum_{r=0}^\infty \left(\frac{t}{\theta}\right)^r , \theta > t. \end{aligned}$$

$$\therefore h'_r = E(X^r) = \text{coefficient of } \frac{t^r}{r!} \text{ in } M_X(t) = \frac{r!}{\theta^r} ; r=1, 2, \dots$$

$$h'_1 = \frac{1}{\theta} , h'_2 = \frac{2}{\theta^2} , h_2 = h'_2 - (h'_1)^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

Hence if $X \sim \exp(\theta)$, Mean = $\frac{1}{\theta}$ and variance = $\frac{1}{\theta^2}$

Remark Variance = $\frac{1}{\theta^2} = \frac{1}{\theta} \cdot \frac{1}{\theta} = \frac{\text{Mean}}{\theta}$

$\therefore$ Variance $>$ Mean, if $0 < \theta < 1$

Variance = Mean if $\theta = 1$

Variance $<$ Mean if $\theta > 1$

Hence for exponential distⁿ Variance $\geq$ or $<$ Mean, for different values of the parameter.

Remark: (i) If $X_1, X_2, \dots, X_n$ are independent r.v.s having exponential distⁿ with parameter $\theta_i ; (i=1, 2, \dots, n)$; then $Z = \min(X_1, X_2, \dots, X_n)$ has exponential distⁿ with parameter $\sum_{i=1}^n \theta_i$.

(ii) If $X_i ; i=1, 2, \dots, n$ are i.i.d following exponential distⁿ with parameter θ , then $Z = \min(X_1, X_2, \dots, X_n)$ is also exponentially distributed with parameter $n\theta$.

If X and Y are two normally correlated variables with correlation coefficient ρ and means and variances (μ_1, μ_2) and (σ_1^2, σ_2^2) respectively. Then the prob distⁿ of bivariate normal distⁿ is given as

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\left[\frac{1}{2(1-\rho^2)}\left\{\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}\right\}\right]}$$

$$\begin{aligned} &-\infty < x < \infty, -\infty < y < \infty \\ &-\infty < \mu_1 < \infty, -\infty < \mu_2 < \infty \\ &\sigma_1^2 > 0, \sigma_2^2 > 0 \end{aligned}$$

$\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and ρ are the parameters of bivariate normal distⁿ

Remark: $(X, Y) \sim BVN(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. If in particular

$$\mu_1 = \mu_2 = 0, \sigma_1 = \sigma_2 = 1, \text{ then } (X, Y) \sim N(0, 0, 1, 1, \rho) \sim BVN(0, 0, 1, 1, \rho)$$

MGF: $M_{X,Y}(t_1, t_2) = E[e^{t_1X + t_2Y}] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{(t_1x + t_2y)} f(x, y) dx dy$

$$= \exp\left[t_1\mu_1 + t_2\mu_2 + \frac{1}{2}(t_1^2\sigma_1^2 + t_2^2\sigma_2^2 + 2\rho t_1t_2\sigma_1\sigma_2)\right]$$

In particular if $(X, Y) \sim BVN(0, 0, 1, 1, \rho)$, then

$$M_{X,Y}(t_1, t_2) = \exp\left[\frac{1}{2}(t_1^2 + t_2^2 + 2\rho t_1t_2)\right]$$

* More about exponential prob distⁿ (Continuation from page 2)

$$f(x, 0) = \theta e^{-\theta x}; x \geq 0, M_X(t) = \left(1 - \frac{t}{\theta}\right)^{-1}$$

$$\text{Mean } \mu_1' = \left.\frac{\partial M_X(t)}{\partial t}\right|_{t=0}; \frac{\partial M_X(t)}{\partial t} = (-1)\left(1 - \frac{t}{\theta}\right)^{-2}\left(-\frac{1}{\theta}\right)$$

$$\mu_1' = \left.\frac{\partial M_X(t)}{\partial t}\right|_{t=0} = \frac{1}{\theta}, \mu_2' = \left.\frac{\partial^2 M_X(t)}{\partial t^2}\right|_{t=0}$$

$$\frac{\partial^2 M_X(t)}{\partial t^2} = (-1)(-2)\left(1 - \frac{t}{\theta}\right)^{-3}\left(-\frac{1}{\theta}\right)\left(-\frac{1}{\theta}\right)$$

$$\mu_2' = \left.\frac{\partial^2 M_X(t)}{\partial t^2}\right|_{t=0} = \frac{2}{\theta^2}, \text{ hence variance } (\mu_2) = \mu_2' - \mu_1'^2 = \frac{1}{\theta^2}$$

If $f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}} ; x > 0$

$$M_X(t) = E e^{tx} = \int_0^{\infty} e^{tx} \frac{1}{\theta} e^{-x/\theta} dx$$

$$= \frac{1}{\theta} \int_0^{\infty} e^{-(\frac{1}{\theta} - t)x} dx$$

$$= \frac{1}{\theta} \int_0^{\infty} e^{-\frac{(1-\theta t)}{\theta} x} dx$$

$$= \frac{1}{\theta} \left[-e^{-\frac{(1-\theta t)}{\theta} x} \cdot \frac{\theta}{(1-\theta t)} \right]_0^{\infty}$$

$$= \frac{1}{\theta} \cdot \frac{\theta}{(1-\theta t)} = \frac{1}{(1-\theta t)} = (1-\theta t)^{-1}$$

$$M_X(t) = (1-\theta t)^{-1}$$

Mean $h_1' = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0} ; \frac{\partial M_X(t)}{\partial t} = (-1)(1-\theta t)^{-2}(-\theta)$

$$h_1' = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0} = \theta$$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0}$$

Hence, $\frac{\partial^2 M_X(t)}{\partial t^2} = (-1)(-2)(1-\theta t)^{-3}(-\theta)(-\theta)$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0} = 2\theta^2$$

$\therefore$ Variance $h_2 = h_2' - (h_1')^2 = 2\theta^2 - \theta^2 = \theta^2$

Mean = θ , Variance = θ^2
